Diego Bravo

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Personal Information

Date of Birth **October 4, 1991**.

Place of Birth **Málaga (Santander)**, *Colombia*.

Citizenship **Colombiano**.

Education

2023-Present **Ph.D. Student in Electrical Engineering**.
University: Universidad Nacional de Colombia, Bogotá, Colombia.

2017-2020 **M.Sc. in Biomedical Engineering**.
University: Universidad Nacional de Colombia, Bogotá, Colombia.

2009-2014 **Mechatronic Engineering**, *University: Universidad Militar, Bogotá, Colombia*.

Awards & recognitions

Distinction **Best Paper Award**, *Congreso Nacional de Gastroenterologia*, Asociación Colombiana de Asociaciones del Aparato Digestivo, November 4, 2023. Bogotá, Colombia.

Distinction **Honorable mention (Mención meritoria)**, *Master's degree in Biomedical Engineering.*, presented in School of Medicine at Universidad Nacional de Colombia, November 3, 2020. Bogotá, Colombia.

Distinction **Best Poster Award**, *Congreso Nacional de Gastroenterologia*, Asociación Colombiana de Gastroenterología, January 8, 2020. Bogotá, Colombia.

Distinction **M.Sc. in Biomedical Engineering**, *The best grades average*, University: Universidad Nacional de Colombia, 2017-2018.

Research Experience

2019-Present **Research Assistant**, *at Universidad Nacional, Director: Eduardo Romero, MD. Ph.D.*
Computer Imaging and Medical Applications Laboratory (Cimalab).

2021-2022 **Research student**, *at Universidad Nacional de Colombia*, Bogotá, Colombia.

2020-2022 **Research**, Departamento Administrativo Nacional de Estadística (DANE), Colombia.

Work Experience

2023-Present **Research Assistant**, *at Universidad Nacional de Colombia*, Bogotá, Colombia.

2021-2022 **Assistant Professor**, *at Universidad Militar*, Bogotá, Colombia.

2014-2017 **Engineer**, *at Pharmaceutical Laboratory*, Bogotá, Colombia.

2014-2014 **Research Assistant - Undergrade**, *at Universidad Militar*, Bogotá, Colombia.

Publications

Journal **Diego Bravo,** Juan Frias, Felipe Vera, et.al., *GastroHUN an Endoscopy Dataset of Complete Systematic Screening Protocol for the Stomach*, Scientific Data - Nature, Nature Publishing Group UK London, 2025.

Conference **Diego Bravo,** Sebastian Medina, Josué Ruano, et.al., *Automatic Endoscopy Classification by Fusing Depth Estimations and Image Information*, ISBI2024, Athens, Greece, May 2024.

Conference Josué Ruano, **Diego Bravo**, Diana Giraldo, et.al., *Estimating Polyp Size From a Single Colonoscopy Image Using a Shape-From-Shading Model*, ISBI2024, Athens, Greece, May 2024.

Conference Maria Jaramillo, **Diego Bravo**, Josué Ruano, et.al., *Predicting Metaplasia in Upper Gastrointestinal Images from White Light Endoscopy*, ISBI2024, Athens, Greece, May 2024.

Conference **Diego Bravo,** Sebastian Medina, Josué Ruano, et.al., *Automated Anatomical Classification and Quality Assessment of Endoscopy by Temporal-Spatial Analysis*, SIPAIM2023, Mexico City, Mexico, November 2023.

Conference Josué Ruano, **Diego Bravo**, María Jaramillo, et.al., *Generating synthetic endoscopy videos following a systematic screening protocol*, SIPAIM2023, Mexico City, Mexico, November 2023.

Conference María Jaramillo, Josué Ruano, **Diego Bravo**, et.al., *Automatic Localization of Pancreatic Tumoral Regions in Whole Sequences of Echoendoscopy Procedures*, SIPAIM2023, Mexico City, Mexico, November 2023.

Conference **Diego Bravo,** Josué Ruano, María Jaramillo,et.al., *Automatic Classification of Esophagogastroduodenoscopy Sub-Anatomical Regions*, ISBI2023, Cartagena, Colombia, April 2023.

Abstract Fabian Emura, **Diego Bravo**, Josué Ruano, et.al., *ID: 3526254 A Novel Deep Learning Model to Facilitate Complete Systematic Photodocumentation During Upper GI Endoscopy*, Gastrointestinal Endoscopy, May 2021.

Conference **Diego Bravo,** Josué Ruano, Martín Gómez, Eduardo Romero, *Automatic polyp detection and localization during colonoscopy using convolutional neural networks*, SPIE Medical Imaging 2022, Houston, Texas, United States, February 2020.

Conference **Diego Bravo,** Josué Ruano, Martín Gómez, et. al., *Automatic polyp localization by low level superpixel features*, SIPAIM2019, Medelin, Colombia, October 2019.

Conference Josué Ruano, **Diego Bravo**, et.al., *Localization of Small Neoplastic Lesions in Colonoscopy by Estimating Edge, Texture and Motion Saliency*, EMBC2019, Berlin, Germany, July 2019.

Conference **Diego Bravo, et. al.**, *Automatic detection of colorectal polyps larger than 5 mm during colonoscopy procedures using visual descriptors*, SIPAIM2018, Mazatlan, Mexico, October 2018.

Research Interests

- Applications in medical imaging and signal processing. Machine Learning, Deep Leaning, Self-supervised learning, Transformers in AI and Ethical AI. Computer Vision and Pattern Recognition. Computer-Aided Diagnosis (CAD) System. Interdisciplinary Collaboration in Healthcare Technology.

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